
Testing Weight-Based Direction Rankings in a Bilinear Robot Policy

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Can a robot policy’s weights identify internal information it can lose while preserv-
2 ing its actions? A reliable ranking could guide analysis and simplification without
3 examples to select each change. We apply an existing tensor decomposition to
4 parts of one trained policy that combines images, instructions, and robot state. Its
5 bilinear layers multiply pairs of transformed vectors. The decomposition ranks
6 internal directions, each a weighted combination of coordinates. Across four re-
7 ductions of one local computation, three random controls per size produce 31 to 53
8 times the leading directions’ action error on 160 recorded inputs. This advantage
9 does not establish better task success. On 20 paired simulator starts, the original
10 policy succeeds on 17, a leading-direction reduction on 16, and the tested random
11 reduction on 19. The difference is inconclusive. A separate vision-block test finds
12 greater output error despite a smaller coefficient-based removal score. We therefore
13 evaluate coefficient scores, action agreement, and task success separately. The
14 evidence covers one checkpoint and limited controls.

15 1 Introduction

16 Robot policies predict actions from images, instructions, and state [4]. Their weights specify which
17 input combinations affect actions. Our question is: **how far does a ranking derived from weights**
18 **carry through to action agreement and task success?** A direction means a weighted combination of
19 an internal vector’s coordinates. We test rankings by removing low-ranked directions and measuring
20 the effect.

21 Bilinear layers multiply pairs of transformed vectors. Pearce et al. [2] use their coefficient tensors for
22 weight-based analysis. Dooms et al. [1] introduce Orthogonalization, Diagonalization, and Truncation
23 (ODT) for χ -nets, networks built from repeated bilinear layers. ODT changes internal coordinates,
24 ranks them using downstream coefficients, and removes selected directions. Their model already
25 reuses inputs and weights. Its normalization becomes fixed after training, and its error bound concerns
26 coefficients. They leave a bound on prediction loss open.

27 We adapt ODT to shared computations and input-dependent rational normalization in one policy. A
28 copied implementation checks every full coordinate change. Local policy reductions and a small
29 vision-block experiment test the gap between its objective and behavior:

- 30 • Full-rank decomposition preserves the tested component outputs to floating-point precision
31 (Section 4).
- 32 • Leading directions preserve recorded actions more closely than the tested random controls,
33 while a 20-start rollout pilot establishes no success advantage (Table 1 and Figure 2).
- 34 • At a large graph-bond budget, spectral allocation incurs greater real-input output error than
35 a width control, despite a smaller coefficient-tail score (Table 2).

Relation to existing methods. Pruning removes weights while controlling prediction error. Optimal Brain Compression and SparseGPT use second-order information for reconstruction [6, 7], while Wanda scores weights using magnitudes and input activations [8]. Tensor-train compression stores a weight tensor as a sequence of smaller cores and measures parameter savings [9]. These are relevant comparators for practical compression. Activation patching replaces internal values to test their effect [10], and distributed alignment search learns directions tied to specified causal variables [11]. Our removals test output sensitivity without assigning semantic meanings. The study follows the broader use of weights as research data [12]. Width and random controls isolate allocation and basis choices. They do not establish superiority to these methods.

2 The bilinear policy and its tensor representation

Our trained policy, χ -VLA, uses the image, instruction, state, and action organization of a vision-language-action model. Its attention and normalization differ from standard transformers. Bilinear attention replaces softmax. Rational normalization was used during training in place of inverse-square-root normalization. The decomposition represents these deployed formulas exactly. General softmax and RMSNorm do not have exact finite polynomial or rational representations of this form.

Architecture. A 64×64 image becomes 64 patch vectors of width 192, called *tokens*, through learned affine maps and position vectors. An affine map is a matrix multiplication plus a bias. Four vision blocks use eight parallel attention heads and 576 feed-forward components each. A projection converts the visual tokens to width 384. The model joins the visual tokens with instruction embeddings, vectors indexed by instruction tokens, a projection of the eight-coordinate state, learned context tokens, and eight action queries. Eight joint blocks use 12 heads and 1,152 feed-forward components. A final normalization and linear head predict eight actions with seven coordinates each. Figure 1 marks the analyzed components.

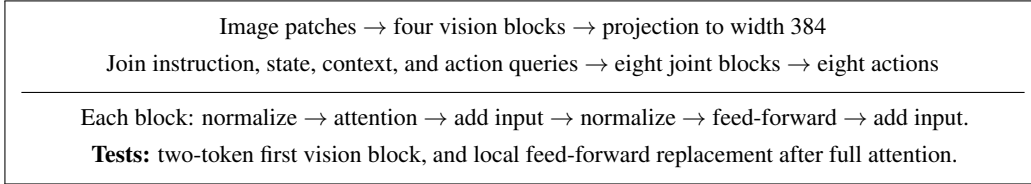


Figure 1: The policy and the two intervention scopes. Each input addition is a residual connection. Only the local feed-forward experiment preserves the original attention context.

Bilinear feed-forward map. For vectors u, v , append a constant coordinate to obtain $\tilde{u} = (u^\top, 1)^\top$ and $\tilde{v} = (v^\top, 1)^\top$. Then

$$B(u, v) = W_D[(W_L \tilde{u}) \odot (W_R \tilde{v})], \quad B_o(u, v) = \sum_{a,b} T_{oab} \tilde{u}_a \tilde{v}_b. \quad (1)$$

Each learned input map W_L, W_R produces 576 components and includes a bias through the constant coordinate. The symbol $\odot$ multiplies matching components, and W_D maps them to the output with no added bias. Expanding gives the coefficient array T . The map is linear in either lifted input with the other fixed. Supplying the same normalized token to both gives a quadratic map.

Attention and residual paths. For head width h , learned maps produce two queries $q_i^{(1)}, q_i^{(2)}$, two keys $k_j^{(1)}, k_j^{(2)}$, and a value v_j . Queries and keys are normalized. With fixed binary mask M and $n_i = \sum_j M_{ij}$, one head computes

$$a_i = \frac{1}{\sqrt{n_i}} \sum_j M_{ij} \frac{(q_i^{(1)\top} k_j^{(1)})(q_i^{(2)\top} k_j^{(2)})}{h^2} v_j. \quad (2)$$

Vision tokens read all image tokens. Joint tokens read current and earlier positions. An affine map combines heads into attention output A . With normalization maps $\mathcal{N}_a, \mathcal{N}_f$ and fixed branch scales α, β , a block applies

$$z = x + \alpha A(\mathcal{N}_a(x)), \quad F(z) = z + \beta B(\mathcal{N}_f(z), \mathcal{N}_f(z)). \quad (3)$$

71 **Rational normalization.** The trained and deployed normalization uses a Padé approximation, a
 72 ratio of polynomials that approximates inverse-square-root scaling:

$$t = \frac{d^{-1} \sum_j x_j^2 + \epsilon}{s}, \quad \mathcal{N}(x) = \frac{x}{\sqrt{s}} \frac{p_0 + p_1 t + p_2 t^2}{q_0 + q_1 t + q_2 t^2}. \quad (4)$$

73 Here d is width, $\epsilon = 10^{-6}$, $s > 0$ is a frozen running mean-square scale, and p_i, q_i are fixed
 74 checkpoint coefficients. The decomposition introduces no further approximation to this formula.

75 **Tensor representation.** A tensor network connects coefficient arrays and sums their shared indices.
 76 An internal index is a *bond*. We represent a rational value by a numerator vector N and denominator
 77 D , decoded as N/D . These *projective coordinates* express the arithmetic with polynomial tensors. A
 78 stored tensor is a *core*. The graph shares cores whose outputs are reused. Copying every use produces
 79 a tree with a separate ordered index for each input occurrence. In inference, copies of the same input
 80 receive equal values. ODT analyzes the coefficient tensor of this fixed representation.

81 3 ODT with a shared basis at every use

82 **Orthogonalization.** We process cores from inputs to output. Put the output index on rows of a
 83 matrix M and the input indices on columns. Direct reduced QR on M^T gives $M = RQ$, where the
 84 rows of Q are mutually perpendicular unit vectors. The factor R is triangular. Store Q and push R
 85 into every parent-input slot, including both slots of a repeated child. The final head receives the root
 86 factor. Dependent rows retain the complete reduced QR basis.

87 Before this step, a core with identical children is averaged over its two input orders. QR uses
 88 symmetric coordinates, with each diagonal entry stored once and each off-diagonal pair weighted by
 89 $\sqrt{2}$. This fixes the coefficient representation. All later full coordinate changes preserve it in exact
 90 arithmetic. Orthogonalization uses direct QR/RQ with no fallback factorization.

91 **Diagonalization and its objective.** For copied occurrence p , explicitly contract two copies of the
 92 downstream network, matching output and other open indices, leaving the cut coordinates free. The
 93 resulting symmetric matrix E_p is its *environment*. For shared bond v , sum over occurrences $\mathcal{O}(v)$
 94 and solve

$$H_v = \sum_{p \in \mathcal{O}(v)} E_p, \quad H_v u_{vi} = \lambda_{vi} u_{vi}, \quad \lambda_{v1} \geq \lambda_{v2} \geq \dots \quad (5)$$

95 An eigenvector u_{vi} is a unit direction scaled by H_v , with scale λ_{vi} . Environments terminate at the
 96 analyzed component's output.

97 Fix its coefficient tensor T and assume the producing subtree has orthonormal output rows after
 98 orthogonalization. Let $T_{p,U}$ restrict only occurrence p to a k -dimensional space with orthonormal
 99 basis vectors $u_1, \dots, u_k$. Then

$$\sum_p \|T - T_{p,U}\|_F^2 = \text{tr}(H_v) - \sum_{i=1}^k u_i^T H_v u_i, \quad \min_U \sum_p \|T - T_{p,U}\|_F^2 = \sum_{i>k} \lambda_{vi}. \quad (6)$$

100 The squared Frobenius norm sums squared entries, and trace sums diagonal entries. Orthogonality
 101 makes removed and retained contributions perpendicular, giving the first equality. The eigenvalue
 102 variational principle maximizes the retained sum with the leading k eigenvectors, giving the second.
 103 This objective adds *separate single-occurrence* errors. Cutting all uses together introduces interac-
 104 tions whose coefficient error is unmeasured here. In Dooms et al.'s symmetric chain, occurrence
 105 environments coincide and their sum rescales the same eigenspaces. We use the stated summed
 106 objective for general sharing.

107 **Truncation.** A full basis transpose enters the producing core, and the basis enters every parent slot.
 108 This preserves the tensor. Retaining only the leading k columns removes directions, with k called the
 109 retained *rank*. We save and reload the smaller cores and check their outputs against masks applied
 110 independently in the full-sized graph. Weights determine the bases. Later policy computations are
 111 used to evaluate their action effects.

112 4 Evaluation protocol

113 **Training and inputs.** We study one 20,137,352-parameter checkpoint trained from scratch on
 114 LIBERO-Object [5], with seed 0, 40,000 steps, batch size 256, learning rate 8×10^{-4} , and 63,352
 115 usable action-chunk examples from 66,984 cached frames. Confirmation uses 80 episodes across
 116 ten tasks and two frames per episode, giving 160 images. Twenty disjoint development episodes
 117 give 40 images. Both partitions come from the training cache. These tests measure agreement on
 118 training-source inputs. They establish no generalization to held-out demonstrations or separately
 119 trained policies.

120 **Components and controls.** The residual feed-forward map F in Equation 3 has six cores and 21
 121 copied occurrences. It replaces the first vision block’s local map after its original 64-token attention,
 122 separately at every token. We reduce one 193-coordinate bond to ranks 192, 184, 174, and 154. Three
 123 random controls per rank use direct QR with seeds 0, 1, and 2. Each retains the bond’s zero-input
 124 activation direction as an *anchor*. This preserves a reference direction, while leaving the remaining
 125 space random. Its effect relative to unanchored random spaces is untested.

126 The two-token block has 453 cores, 81,377 copied occurrences, and 9,274 eligible directions across
 127 450 bonds. Eligible bonds exclude inputs and the final output, and each retains at least one direc-
 128 tion. Two tokens make the copied correctness check tractable. Removing 62 tokens itself causes
 129 RMSE 0.2753, or 18.2% of full-context output RMS. Block reductions therefore use the unchanged
 130 two-token output as reference. Real inputs are patch pairs (27, 28) and (0, 63) from each image,
 131 giving 320 inputs. The 256 synthetic inputs have independent Gaussian entries with mean zero and
 132 standard deviation 0.2, using seed 20260907. These panels have different scales and structure. Their
 133 comparison cannot isolate an input-distribution effect.

134 **Width allocation (W)** chooses removals by increasing $(2j - 1)/(2d_v)$, for width d_v and removal
 135 count j . This is the incremental cost of $j^2/(2d_v)$ and encourages similar fractional cuts. **Spectral**
 136 **allocation (S)** removes the smallest tail eigenvalues across bonds, on a shared scale including
 137 occurrence counts. **Normalization protection (N)** keeps 136 mean-square and Padé bonds intact,
 138 yielding WN and SN. Random block controls match all retained ranks at 10% and 30%. Exact
 139 duplicates leave 25 conditions. Additional $K = 1, 9, 46$ cuts are exploratory. All percentages count
 140 graph-bond directions, which include fixed arithmetic coordinates.

141 **Checks and metrics.** An independent implementation copies every use and replays each full
 142 coordinate change. Full-rank scaled output errors are 4.60×10^{-15} for the block and 1.74×10^{-14}
 143 for F . Independently contracted occurrence environments agree to 2.07×10^{-14} . New block cutoffs
 144 use these accepted bases and eigen-equation checks. Independent per-cut copied checks cover the
 145 original producer schedule and all four local feed-forward ranks.

146 Root mean square error (RMSE) takes the square root of mean squared output differences. Block
 147 error is also divided by reference output RMS, 1.3823 on real and 0.6317 on synthetic inputs.
 148 Action error combines all 8×7 denormalized outputs, with a second metric dividing each error
 149 coordinate by its training-target standard deviation. Finite projective coordinates pass decoding when
 150 $|D|/\max(|D|, \max_j |N_j|) > 10^{-12}$. This bounds decoded output size and is not a conditioning
 151 certificate. All block and action conditions complete every input without decoding failure. Paired
 152 action intervals resample 80 episodes within ten tasks 4,000 times. They are descriptive, pointwise
 153 intervals conditional on this checkpoint.

154 5 Results

155 **Local ranks preserve recorded actions.** Full-rank replacement agrees with original actions at
 156 RMSE 7.34×10^{-8} . At each reduced rank, all three anchored random controls have greater RMSE
 157 than the leading basis (Table 1). Their error ratios span 31 to 53 across all twelve comparisons.
 158 Leading-basis median frame errors range from 2.03×10^{-5} to 2.80×10^{-4} , and 95th percentiles
 159 from 4.14×10^{-5} to 5.59×10^{-4} . The worst 5% of frames contribute 19.5% to 23.3% of their
 160 squared error. Every arm preserves all 1,280 reference gripper signs. Appendix A gives individual
 161 controls, tails, and all paired intervals.

Retained rank	Leading RMSE	Leading standardized RMSE	Anchored random RMSE range
192	0.0000244	0.0000990	0.001004 to 0.001300
184	0.0000918	0.0003757	0.003653 to 0.004426
174	0.0001634	0.0006697	0.005913 to 0.006681
154	0.0003269	0.0013681	0.010107 to 0.013384

Table 1: Action agreement on 160 training-source inputs. One 193-coordinate bond is reduced. Standardized errors use training-target standard deviations, with no division by prediction RMS. Random ranges cover three anchored bases. These are controls for direction choice.

Action agreement does not establish a task-success benefit. A pilot evaluates four arms on the same 20 LIBERO-Object starts, two per task, with a 280-step limit, eight-action chunks, and ten settling steps. All arms use the same direct-QR constrained-dynamics controller, simulator versions, and RTX 6000 hardware. All 80 episodes finish without execution failures. Native and full-rank arms have identical success vectors (Figure 2). The random reduction succeeds on three starts where the leading reduction fails, with no discordance in the opposite direction. An exact two-sided paired test gives $p = 0.25$. The pilot establishes neither superiority nor equivalence.

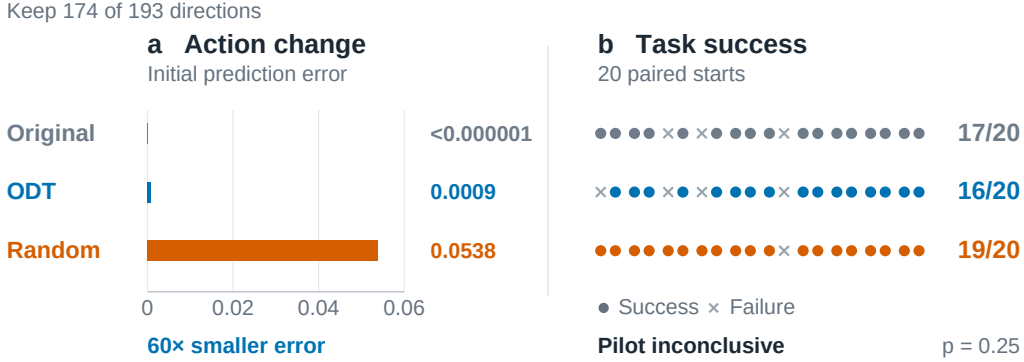


Figure 2: Action agreement and task success on the same 20 starts. Left: first-chunk RMSE relative to full-rank replacement, evaluated post hoc. Right: one column per paired start. ODT and anchored random seed 0 each retain 174 of 193 directions in one feed-forward bond. The paired success comparison is inconclusive ($p = 0.25$).

On those same starting observations, the leading reduction has smaller first-chunk error on all 20 starts. The random arm’s aggregate RMSE is 59.6 times larger. This second input source confirms initial action agreement. It does not establish fidelity later in the rollout or independence from policy training states. Smaller offline error alone therefore provides insufficient evidence for improved task success.

Block allocation has a narrow low-error regime. Table 2 includes the smallest cuts in the main evidence. At $K = 93$, spectral allocation has real-input error of 0.0115% of source RMS. It rises to 2.28% at $K = 464$ and 41.20% at $K = 2782$. Across the spectral schedule, sub-percent relative error is demonstrated only through roughly 1% graph-bond removal. At the largest budget, W has lower real-input error than S. The paired squared-error difference, S minus W, is 0.18379 with descriptive interval [0.17996, 0.18737]. WN improves on W by 12.8%. These control comparisons show a limitation of the coefficient-tail objective. They establish no advantage over calibrated pruning baselines.

Large cuts concentrate in different computations. At $K = 2782$, S cuts 94 wider bonds and leaves all 136 normalization and 96 attention-scalar bonds intact. W cuts all 450 eligible bonds. S reduces both normalized feed-forward inputs from 193 to 43, against W’s 135, and both attention outputs from 193 to 57, against W’s 135. Its summed coefficient-tail score is 146.93, against W’s 478,715.91 in common stored units. This diagnoses different allocations without identifying a single causal bond. S’s denominator magnitude changes by only -0.0272 to 0.0598 bits relative to the fixed reference, with no sign changes. Denominator collapse does not explain this case. At the same budget, the W seed-0 random control has RMSE 537.35, a 95th-percentile input error of 761.44, and

K	Graph-bond removal (%)	Real-input RMSE / source RMS (%)			
		W	WN	S	SN
1	0.0108	0.73	0.73	6.282×10^{-6}	6.282×10^{-6}
9	0.0970	1.32	1.32	3.331×10^{-4}	3.331×10^{-4}
46	0.4960	2.16	2.16	0.0039	0.0039
93	1.0028	3.04	3.04	0.0115	0.0115
464	5.0032	6.29	6.29	2.28	2.28
927	9.9957	7.96	7.96	6.98	6.98
2782	29.9978	27.12	23.66	41.20	41.20

Table 2: Leading directions under width (W) or spectral (S) allocation, with normalization protection (N) shown explicitly. The $K = 1, 9, 46$ extension is exploratory. The denominator is 9,274 eligible graph-bond directions, unrelated to policy parameter count.

190 a maximum of 5,748.31, despite every input passing decoding. Appendix B reports ranks and tails. A
191 decoding pass alone gives no error guarantee.

192 6 Interpretation and limitations

193 The central gap is between coefficients and decoded outputs. If $y = N/D$, then a cut gives

$$\delta y = \frac{D \delta N - N \delta D}{D(D + \delta D)}. \quad (7)$$

194 Cancellation and both denominators affect error. Copied tensor inputs also occupy independent
195 coefficient axes while inference gives them equal values. Nissen Gonzalez et al. [3] construct
196 symmetry-invariant similarities for polynomial model functions using symmetrized tensors and a
197 function-space metric based on averages over Gaussian inputs. Our fixed ordered representation is
198 different. A rational output has equivalent pairs (N, D) and (gN, gD) for common nonzero factors
199 g on the domain. Their polynomial similarity does not directly remove this rational-representation
200 freedom. Our spectra are therefore conditional on the chosen representation.

201 This study demonstrates audited local analysis and an objective-to-behavior gap. It identifies no
202 named features or practical compression gain. Recorded inference times are 19.2 ms per input for
203 the native policy, 44.7 ms for full-rank replacement, and 41.1 ms at rank 154. These are run timings,
204 without repeated benchmarking. Expanded feed-forward core storage falls from 14.56 to 11.66
205 million scalars, compared with 332,928 learned parameters in the original feed-forward matrices and
206 input biases. Graph savings provide no evidence of whole-policy memory or speed savings.

207 Stronger magnitude or activation-based baselines, unanchored random spaces, multiple checkpoints,
208 training-held-out inputs, longer attention contexts, and direct simultaneous-cut coefficient errors
209 remain untested. The Gaussian and image panels differ in both scale and structure, so we draw no
210 causal conclusion from their differing orderings. Larger paired rollouts are needed to test success
211 retention. For this checkpoint, coefficient rankings, action agreement, and task success remain
212 separate evaluation targets.

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A Individual action controls and uncertainty

The frozen action experiment has 18 arms, including the original policy, full rank, four leading ranks, and twelve anchored random controls. Table 3 reports per-frame tails and standardized error. Standardization divides the seven error coordinates by training-target standard deviations (0.27267, 0.44593, 0.45798, 0.024709, 0.049447, 0.042694, 0.99037) before squaring and averaging. It gives rotation, translation, and the continuous gripper output comparable target scales. Raw action RMSE retains their different scales. Neither metric is relative to the RMS of reference predictions.

Rank	Basis	RMSE	Standardized	Median frame	95th percentile	Maximum frame
192	Leading	0.0000244	0.0000990	0.0000203	0.0000414	0.0000739
192	Anchor 0	0.0010310	0.0045519	0.0008413	0.0017697	0.0030800
192	Anchor 1	0.0013001	0.0057482	0.0010789	0.0021564	0.0031260
192	Anchor 2	0.0010043	0.0045471	0.0008123	0.0016854	0.0033655
184	Leading	0.0000918	0.0003757	0.0000798	0.0001502	0.0002338
184	Anchor 0	0.0036526	0.0155959	0.0028930	0.0063518	0.0095541
184	Anchor 1	0.0044265	0.0179281	0.0039240	0.0074330	0.0122404
184	Anchor 2	0.0041050	0.0185172	0.0036749	0.0068004	0.0105014
174	Leading	0.0001634	0.0006697	0.0001377	0.0002604	0.0005451
174	Anchor 0	0.0059129	0.0247959	0.0048050	0.0101447	0.0161449
174	Anchor 1	0.0064158	0.0251769	0.0054522	0.0103743	0.0132616
174	Anchor 2	0.0066815	0.0285241	0.0056302	0.0112339	0.0203611
154	Leading	0.0003269	0.0013681	0.0002804	0.0005591	0.0008305
154	Anchor 0	0.0133840	0.0411569	0.0082934	0.0174267	0.1048555
154	Anchor 1	0.0106916	0.0429834	0.0090762	0.0169847	0.0270213
154	Anchor 2	0.0101069	0.0440125	0.0088848	0.0161733	0.0240049

Table 3: All reduced action arms on the same 160 inputs. Frame summaries use each frame’s RMSE across eight actions and seven coordinates.

Rank	Anchor seed	Paired MSE difference	95% lower	95% upper
192	0	-1.0623×10^{-6}	-1.2334×10^{-6}	-9.1797×10^{-7}
192	1	-1.6896×10^{-6}	-1.9246×10^{-6}	-1.4766×10^{-6}
192	2	-1.0080×10^{-6}	-1.1718×10^{-6}	-8.6774×10^{-7}
184	0	-1.3333×10^{-5}	-1.5205×10^{-5}	-1.1584×10^{-5}
184	1	-1.9585×10^{-5}	-2.2429×10^{-5}	-1.6997×10^{-5}
184	2	-1.6843×10^{-5}	-1.8649×10^{-5}	-1.5141×10^{-5}
174	0	-3.4935×10^{-5}	-4.0123×10^{-5}	-3.0324×10^{-5}
174	1	-4.1135×10^{-5}	-4.5988×10^{-5}	-3.6773×10^{-5}
174	2	-4.4616×10^{-5}	-5.1883×10^{-5}	-3.7820×10^{-5}
154	0	-1.7902×10^{-4}	-3.2806×10^{-4}	-8.5026×10^{-5}
154	1	-1.1420×10^{-4}	-1.2965×10^{-4}	-9.9604×10^{-5}
154	2	-1.0204×10^{-4}	-1.1482×10^{-4}	-8.9555×10^{-5}

Table 4: Leading minus matched random mean squared action error. Whole episodes are resampled within tasks, with 4,000 replicates and seed 2026090703. All intervals lie below zero. They are pointwise descriptive intervals, without multiple-comparison correction.

The leading arms’ largest eight frame errors contribute 19.5% to 23.3% of total squared error. Basis seeds describe retained-space variation within one checkpoint. They provide no estimate of variation between independently trained policies. The action analyzer independently recomputes metrics from saved arrays and verifies source identities. It algebraically verifies archived 64-bit comparison receipts without re-executing GPU inference.

B Block controls, allocation, and decoding diagnostics

All 25 primary conditions evaluate 320 image-derived and 256 Gaussian inputs. Their source output RMS values are 1.3823244 and 0.6316512. Table 5 gives absolute values for every condition. Equivalent schedules share one row. At the 30% graph-bond budget, S has lower error than W on this Gaussian panel and higher error on this image panel. Scale and input structure are both unmatched, so these panels do not identify the cause of the difference.

ID	Allocation	K	Basis	Real RMSE	Synthetic RMSE
0	Full	0	Leading	2.20629×10^{-14}	1.15092×10^{-14}
1	W / WN	93	Leading	0.042030	0.058148
2	W / WN	464	Leading	0.086894	0.106534
3	W / WN	927	Leading	0.109973	0.142223
4	W / WN	927	Anchor seed 0	1.064549	0.573082
5	W / WN	927	Anchor seed 1	1.060635	0.575477
6	W / WN	927	Anchor seed 2	0.952759	0.549312
7	W	2782	Leading	0.374901	0.467858
8	W	2782	Anchor seed 0	537.354468	3.997828
9	W	2782	Anchor seed 1	28.011595	4.190110
10	W	2782	Anchor seed 2	60.514962	3.688361
11	WN	2782	Leading	0.327004	0.334772
12	WN	2782	Anchor seed 0	1.485593	0.742835
13	WN	2782	Anchor seed 1	1.244759	0.731174
14	WN	2782	Anchor seed 2	1.297694	0.730426
15	S / SN	93	Leading	0.000159	0.000164
16	S / SN	464	Leading	0.031470	0.030965
17	S / SN	927	Leading	0.096475	0.127134
18	S / SN	927	Anchor seed 0	1.813258	0.670609
19	S / SN	927	Anchor seed 1	1.038375	0.657134
20	S / SN	927	Anchor seed 2	1.189186	0.651609
21	S / SN	2782	Leading	0.569509	0.333199
22	S / SN	2782	Anchor seed 0	3.221475	0.678179
23	S / SN	2782	Anchor seed 1	1.681646	0.676258
24	S / SN	2782	Anchor seed 2	1.747298	0.676456

Table 5: Complete frozen comparison. Displayed values are rounded. Source RMS values are given above. These controls are conditional on one checkpoint and the fixed two-token representation.

272 **Allocation at $K = 2782$.** W removes 136 directions from all 136 moment/Padé bonds, 96 from all
273 96 attention-scalar bonds, and 2,550 from 218 wider bonds. WN reallocates the first 136 removals
274 to wider bonds. S removes all 2,782 directions from 94 wider bonds. Its remaining 356 bonds stay
275 intact. Table 6 reports retained ranks grouped by computation. This is a post hoc description of the
276 implemented schedules. Isolating the contribution of each cut would require separate interventions.

Bond computation	Original width range	W retained range	S retained range
Mean-square and Padé	2	1	2
First query projection	25	17 to 18	25
First key projection	25	18	25
Second query projection	25	18	25
Second key projection	25	18	25
First query normalization	25	17 to 18	17 to 25
Second query normalization	25	18	16 to 25
First key normalization	25	18	17 to 25
Second key normalization	25	18	16 to 25
Value projection	25	18	6 to 22
Weighted values	25	18	4 to 19
Attention route sum	25	18	6 to 22
Joined head outputs	49 to 193	34 to 135	29 to 57
Block attention input	193	135	193
Attention output	193	135	57
Post-attention residual	193	135	193
Normalized feed-forward input	193	135	43
Feed-forward branch	193	135	72
Post-feed-forward residual	193	135	72

Table 6: Ranges across bonds with the same computational role. Input and final output bonds are excluded. S concentrates reductions after attention and before or within feed-forward computation.

277 **Large error remains possible after a decoding pass.** For finite $y = N/D$, the validity ratio equals
278 $1/\max(1, \max_j |y_j|)$. Its threshold limits output magnitude. The proposed ratio $|D|/\max_j |N_j|$
279 likewise measures reciprocal output magnitude, without certifying numerical conditioning. Absolute
280 denominator values also change under a common projective rescaling. We therefore inspect denom-
281 inator changes relative to the source in the fixed representation, carrying all power-of-two scales
282 explicitly.

283 For S at 30%, $\log_2 |D_{\text{cut}}/D_{\text{source}}|$ has minimum -0.02718 , median 0.02543 , and maximum 0.05979 ,
284 with no sign changes. Its decoding-margin minimum is 0.1101 . In contrast, the W seed-0 random
285 control has 84 denominator sign changes across 320 inputs and a minimum decoding margin of
286 4.09×10^{-5} . Its large RMSE is accompanied by a broad tail (Table 7). Changes in this chosen
287 representation do not isolate conditioning from simultaneous numerator changes.

Allocation	Basis	RMSE	Relative RMSE (%)	95th percentile	Maximum input
W	Leading	0.37490	27.12106	0.44891	0.80824
W	Anchor 0	537.35447	38873.25342	761.44059	5748.31259
W	Anchor 1	28.01159	2026.41251	48.22915	127.39276
W	Anchor 2	60.51496	4377.76849	81.91347	420.75352
WN	Leading	0.32700	23.65608	0.43956	0.55265
WN	Anchor 0	1.48559	107.47065	1.80696	1.93944
WN	Anchor 1	1.24476	90.04829	1.76630	1.90639
WN	Anchor 2	1.29769	93.87765	1.73215	1.96965
S / SN	Leading	0.56951	41.19937	0.72477	0.77833
S / SN	Anchor 0	3.22148	233.04771	5.09337	21.98588
S / SN	Anchor 1	1.68165	121.65349	2.43831	2.92744
S / SN	Anchor 2	1.74730	126.40290	2.45517	2.86129

Table 7: Real-input error tails at $K = 2782$. Percentiles and maxima refer to each input’s RMSE across 384 output coordinates. Median errors and full block-error distributions are unavailable in the retained aggregate summaries.

C Correctness, costs, and rollout scope

Independent replay. Full-rank scaled error is $\max |a - b| / \max(\max |b|, 10^{-12})$, with reference b . It is 4.60×10^{-15} for the block and 1.74×10^{-14} for the local feed-forward map. Copied every-step core comparisons agree exactly at recorded floating-point entries. Independent occurrence environments agree to 2.07×10^{-14} . The block producer checked 1,540 retained spaces in its original schedule. Additional campaign cutoffs use accepted bases, eigen-equation checks, and reduced-core versus full-mask replay. They were not independently copied and checked at each new cutoff. The local producer directly checked its four reduced ranks. Full-rank native action error is 7.34×10^{-8} in denormalized units.

Context and arithmetic. The original 32-bit calculation and an independently written 64-bit forward have maximum absolute differences 4.69×10^{-6} for the two-token block and 8.04×10^{-6} for the 64-token block. A full-context calculation masked to the selected pair agrees with the two-token computation to 3.81×10^{-6} . The 18.2% context-removal error uses full-context output RMS 1.5156. Every truncation result instead uses the two-token reference. No 8-token or 16-token decomposition was tested.

Recorded costs. Local decomposition took 34.6 seconds. Preparing the block campaign from accepted bases took 206.3 seconds, including 1,350 QR calls for anchored bases. Original block decomposition, peak memory, and context-length scaling costs are unavailable. Table 8 counts serialized core entries, excluding the unchanged output head and saved bases. Inference timings are single-run full-policy means. They show no acceleration over native inference.

Representation	Stored core scalars	Full-policy ms / input
Native policy	Not a core representation	19.22
Local map, full rank	14,564,367	44.67
Local map, rank 174	13,148,905	43.18
Local map, rank 154	11,658,945	41.10
Two-token block, full rank	64,051,337	Not measured
Two-token block, W at $K = 2782$	26,299,858	Not measured
Two-token block, S at $K = 2782$	10,203,600	Not measured

Table 8: Representation storage and recorded run timings. Core counts include fixed arithmetic, input cores, and numerator-denominator coordinates. They are not parameter counts or peak-memory measurements.

Rollout implementation. All arms share MuJoCo 3.5.0, robosuite 1.4.1, LIBERO 0.1.0, PyTorch 2.7.1, and Quadro RTX 6000 GPUs. The image is rotated by 180° , resized to 64×64 , and divided by 255. The sign of the predicted gripper coordinate determines its discrete action. A direct Householder-QR solver handles the constrained-dynamics controller, with no fallback for singular systems. Its largest normwise backward error, the linear-system residual divided by the stated matrix and solution scale, is 1.48×10^{-17} against a 10^{-12} gate. Its equivalence to the historical controller is restricted to nonsingular systems. These results should be interpreted under this controller and horizon.

Descriptive Wilson 95% intervals for success are 64.0% to 94.8% for native and full rank, 58.4% to 91.9% for leading rank 174, and 76.4% to 99.1% for the random arm. They summarize this small balanced pilot. The exact paired comparison between leading and random has three discordances and $p = 0.25$. A within-task bootstrap with only two starts per task is too fragile to support a superiority claim. Equal native and full-rank success vectors provide a sanity check without establishing statistical equivalence.

Evidence availability. We retain protocols, source snapshots, input identities, output summaries, action errors, and rollout outcomes. Internal hashes detect artifact changes. This draft provides no anonymous link to the complete checkpoint and raw predictions, limiting external reproduction.